Summary in Graph

Exam Summary (GO Classes GATE CS | Theory of Computation | FA, CFL, DCFL | Weekly Quiz 5)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	10 (5 + 5)
Total Marks:	15 (5 + 10)
Exam Duration:	30 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained

Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10

* Average Stats based on first-attempt stats of Top 10% (28) out of 284 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Choice Type Award: 1 Penalty: 0.33 Theory of Computation

In the CFG $G = S \longrightarrow 0S|1S|\epsilon$:

A. The string 011 is ambiguously derived;
B. The string 010 is ambiguously derived;
C. The empty string ϵ is ambiguously derived;
D. No string in $L(G)$ is ambiguously derived.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

If two different variables in a context-free grammar G each derive ϵ, then:

A. G must be ambiguous

B. $\epsilon \in L(G)$

C. Converting G to G' in Chomsky normal form makes $L(G') \neq L(G)$

D. None of the above necessarily happens.

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

The concatenation $\emptyset \cdot \Sigma^*$ equals:

A. $\emptyset$;
B. $\{\epsilon\}$;
C. Σ^*;
D. None of the above.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Theory of Computation

In the CFG $G = S \longrightarrow aS|Sa|b$,

A. The string aba is ambiguously derived;
B. The string aab is ambiguously derived;
C. The string bb is ambiguously derived;
D. No string in $L(G)$ is ambiguously derived.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 01sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Theory of Computation

Which of the following statements is/are TRUE?

A. Every finite language is DFA-acceptable.
B. For all L, L^+ is infinite.
C. If $M = (Q, \Sigma, \delta, q_0, F)$ is a DFA and F = Q, then $L(M) = \Sigma^*$
D. If $M = (Q, \Sigma, \delta, q_0, F)$ is a NFA and F = Q, then $L(M) = \Sigma^*$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

The union of two non-regular DCFLs can possibly be (mark all that are possible):

A. Regular
B. A non-regular DCFL
C. A CFL that is not a DCFL
D. Non-CFL

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

For every language A, the concatenation A^*A^* equals:

A. $(AA)^*$
B. Σ^*
C. A^*
D. None of the above.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Theory of Computation

If M_1, M_2, and M_3 are DFAs with 100 states each and the same alphabet Σ, then $L(M_1) \cup L(M_2) \cup L(M_3)$ is:

A. Always equal to Σ^*;
B. Always recognized by a DFA with 300 states;
C. Always recognized by an NFA with 301 states;
D. Possibly non-regular.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements is/are false?

A. The intersection of a CFL and a DCFL is always a CFL.
B. The union of two regular languages is always a DCFL.
C. For every nondeterministic PDA N, there is a deterministic PDA M such that $L(M) = L(N)$.
D. The intersection of two non-regular languages is always non-regular.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Select Type | Award: 2 | Penalty: 0 | Theory of Computation

Which of the following statements is/are true ?

A. If $A^* = A$, then the language A includes the empty string.
B. If A and B are regular languages recognized by 3 -state DFAs, then $A \cap B$ can be recognized by a 6 -state DFA.

C. Every language generated by an unambiguous context-free grammar is accepted by some DPDA.
D. Every subset of a regular language is regular.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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